Reg. No. | | | | | | | | | | | [MSCCH203]

SECOND SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (CHEMISTRY)

PHYSICAL CHEMISTRY-II

Time : 03 Hours **Maximum Marks:80**

Instructions: **1.** Questions 1 to 6 carry 15 marks each, answer any four.

2. Question 7 is compulsory, carries 20 marks.

1. **a.** Define metastable curves, component, phase and degree of freedom.

b. State the significance of reduced phase rule over phase rule with an example.

c. Derive the equation for phase rule using chemical potentials. **5+5+5**

2. **a.** Explain the terms, galvanic corrosion, cervic corrosion, filiform corrosion, inter-granular corrosion and corrosion fatigue.

b. How formations of anodic and cathodic regions contribute to corrosion? **10+5**

3. **a.** Explain any three types of potentiometric titrations in detail.

b. Give an account on the application of coulometric titrations. **10+5**

4. **a.** Describe autocatalysis and oscillatory reactions.

b. Explain mathematical expression of an auto catalytic reaction.

c. Explain the characteristics of enzyme catalysis. **5+5+5**

5. **a.** Give the classification of carbonaceous materials. List out the desired characteristics of anode material.

b. Explain the silver button battery with applications. **10+5**

6. **a.** Explain the differences between conventional battery and solid-state battery.

b. What is metal/air battery? Give the list of different metal/air batteries.

c. Write a note on alkaline-fuel cell. **5+5+5**

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7. **Answer any TEN of the following.** **10×2=20**

a. What is reduced phase rule?

b. Give any two examples for single component system having three phases.

c. Define dry and wet corrosion.

d. What is the difference between resistance and impedance?

e. What are the principles underlying amperometric titrations?

f. Write any two advantages and disadvantages of amperometric titrations.

g. Define homogeneous and heterogeneous catalysis with an example.

h. What is the role of promoters in catalytic reactions with an example?

i. List out the important components of a battery.

j. Define battery and give the classification of batteries.

k. What is CO_2 poisoning in battery?

l. What is membrane-electrode-assembly?